
Microneedle Microsystems for Glucose Monitoring and Closed-Loop Insulin Delivery

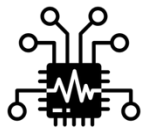
Micro-electromechanical Systems

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Introduction



Development of a closed-loop system integrating continuous glucose monitoring (CGM) and automated insulin delivery



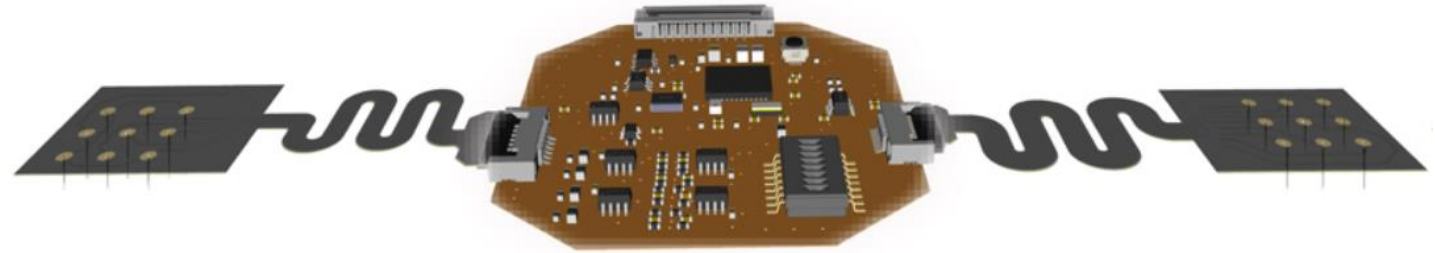
Application of microneedle arrays for minimally invasive interstitial fluid (ISF) sensing and drug release



Step-by-step analysis of the manufacturing processes, design trade-offs, and reliability of four selected microsystems



Functioning and innovation of the Integrated Individually Addressable Microneedle Arrays



Complete Closed-Loop System

Integrates two distinct 3×3 microneedle arrays (for sensing and therapy) via a Bluetooth connection



Redundant Sensing

Four independent electrodes act as parallel glucose sensors, ensuring valid readings even if some needles fail



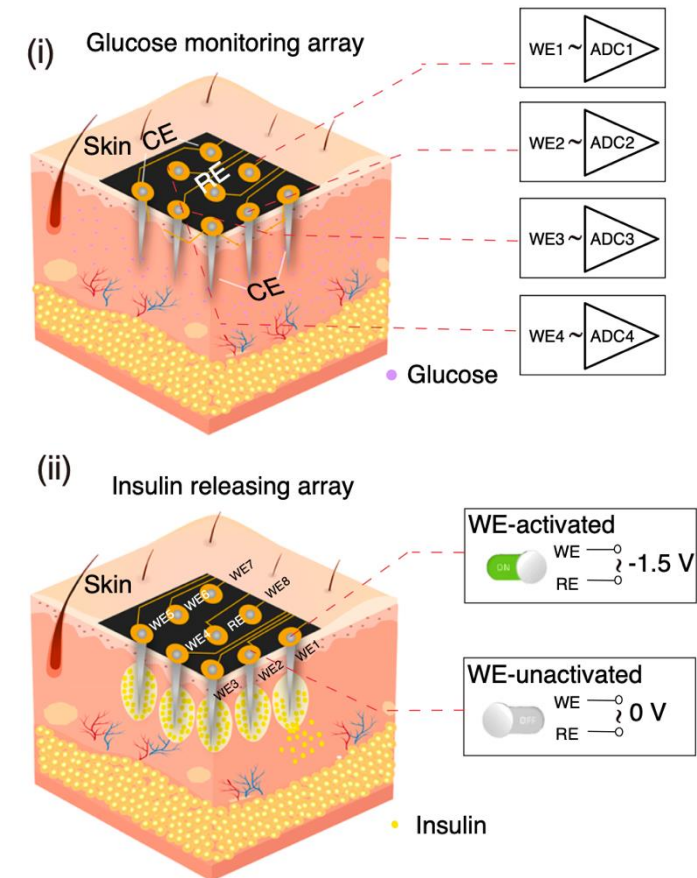
Active Delivery

Replaces passive diffusion with active, electrochemical release, delivering precise 1-unit doses of insulin on demand



Automated Control

Glucose data streams to a smartphone every second, automatically triggering insulin release when thresholds are crossed



Manufacturing



Platinum catalytic layer: electrodeposition



Permselective membrane: electropolymerisation



Enzyme immobilisation: glutaraldehyde cross-linking



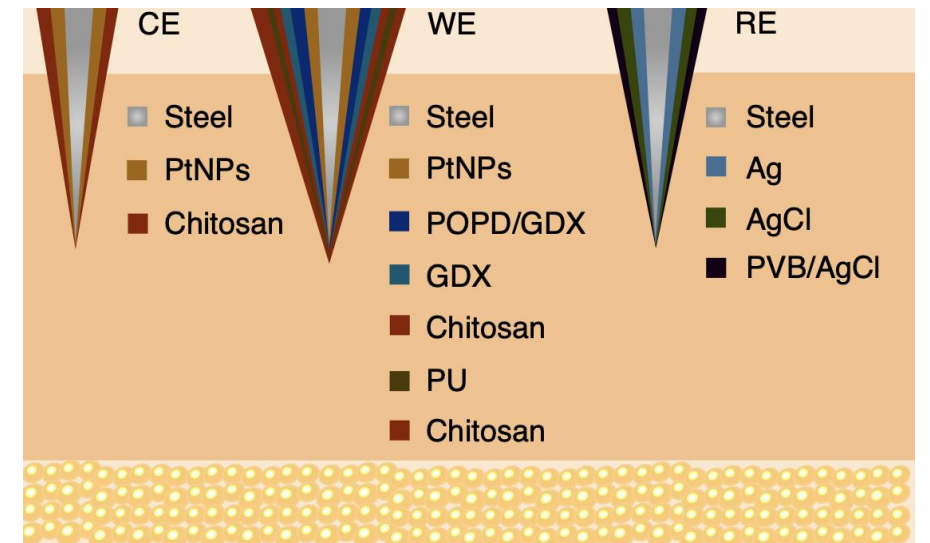
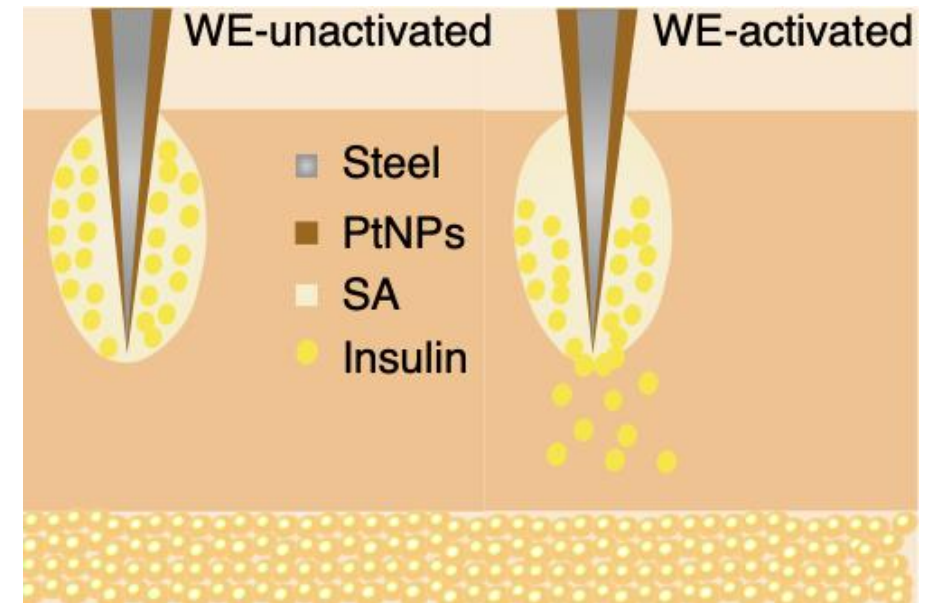
Protective encapsulation: solution coating



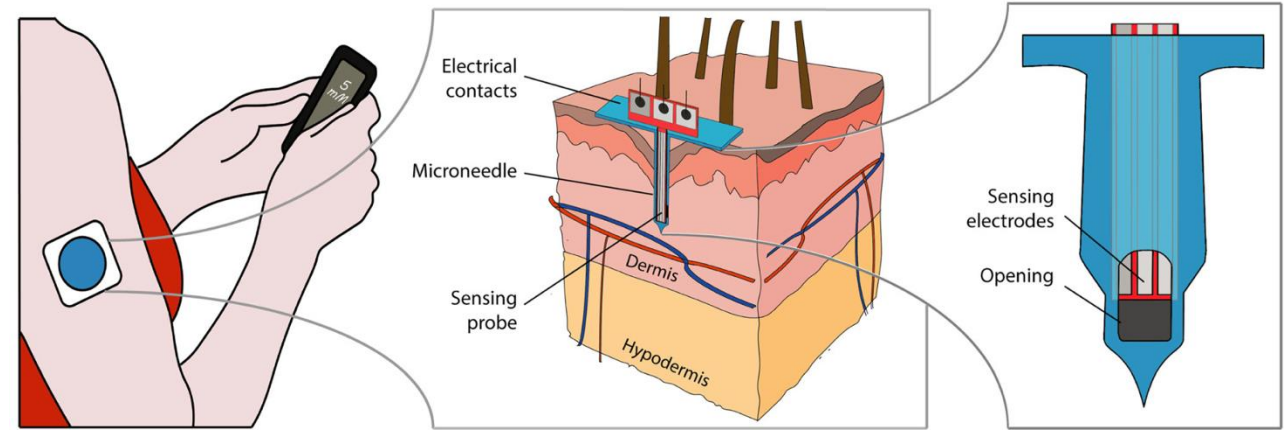
Insulin needles – galvanostatic gel deposition



Array assembly and packaging

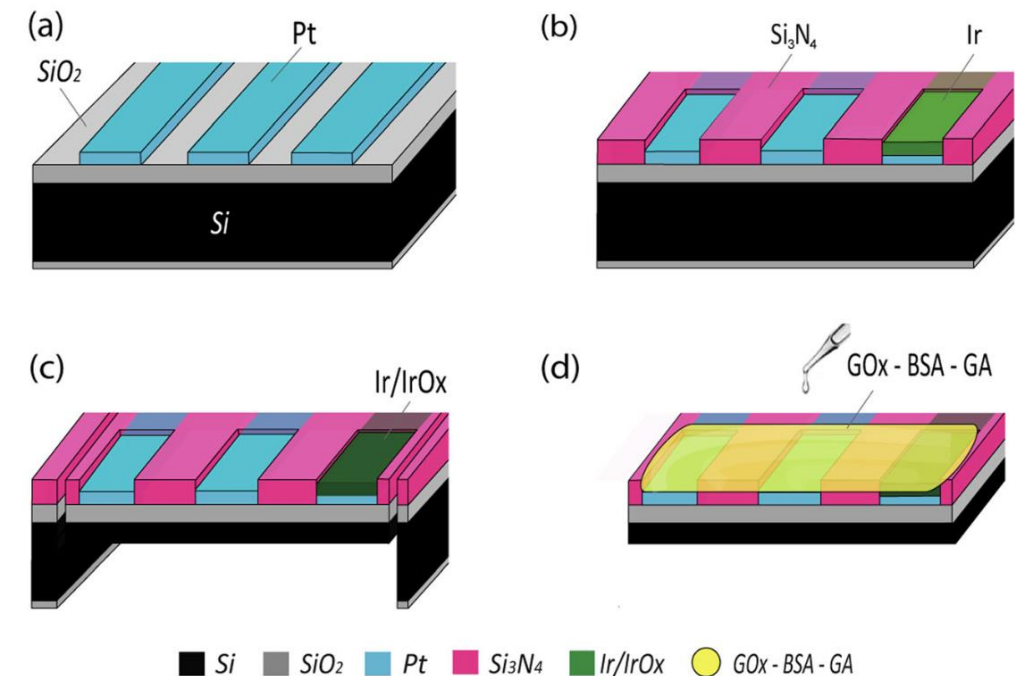


Hollow-Silicons Sensing Needle



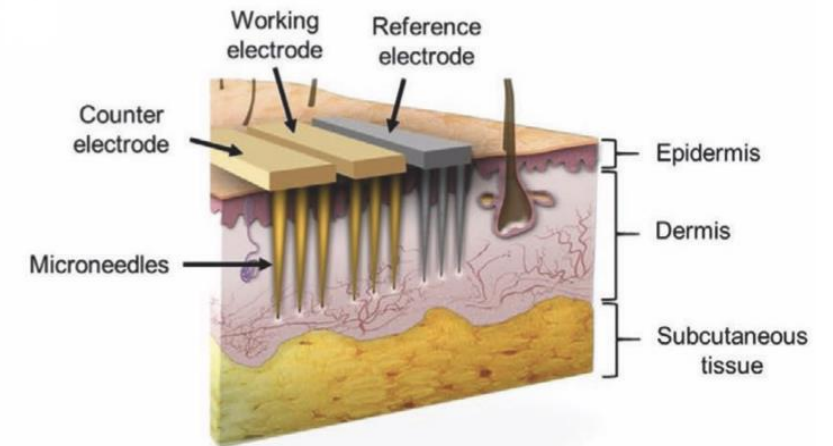
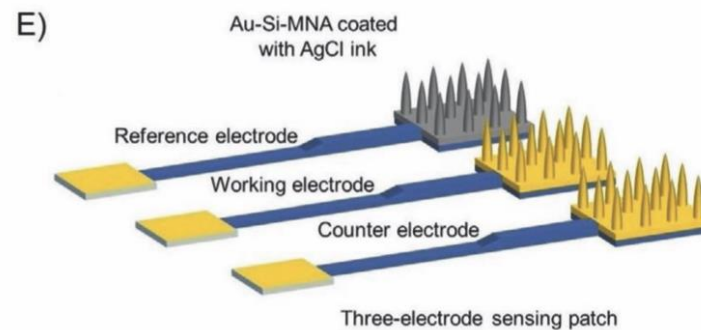
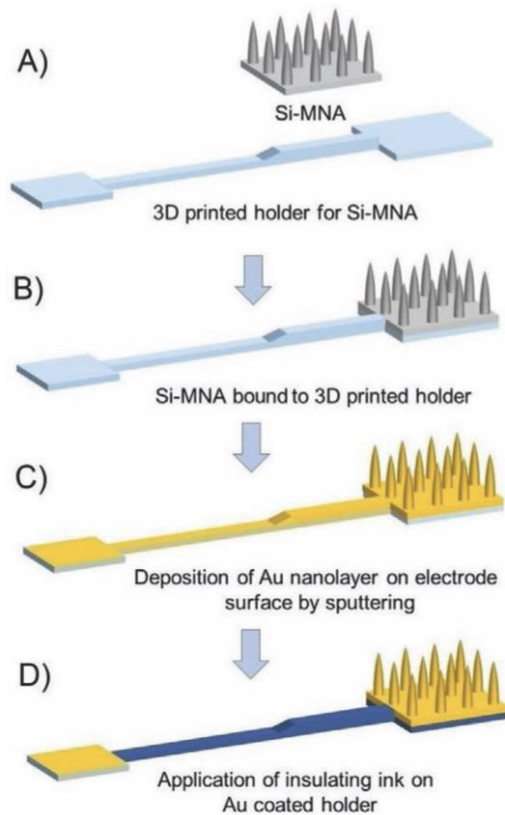
- Single hollow silicon microneedle containing an ultra-miniaturised 3-electrode probe
- Relies on passive capillary action (<1 nL) and diffusion for rapid sensing (~10 min lag)
- Exceptionally small transdermal footprint to minimize trauma and ensure painless insertion

1	Insulation: thermal oxidation
2	Electrodes: photolithography, lift-off and electron-beam evaporation
3	Passivation and openings: PECVD and wet etching
4	Singularisation: deep reactive-ion etching and back-thinning
5	Hollow needle: isotropic and anisotropic dry etching
6	Reference and functionalisation

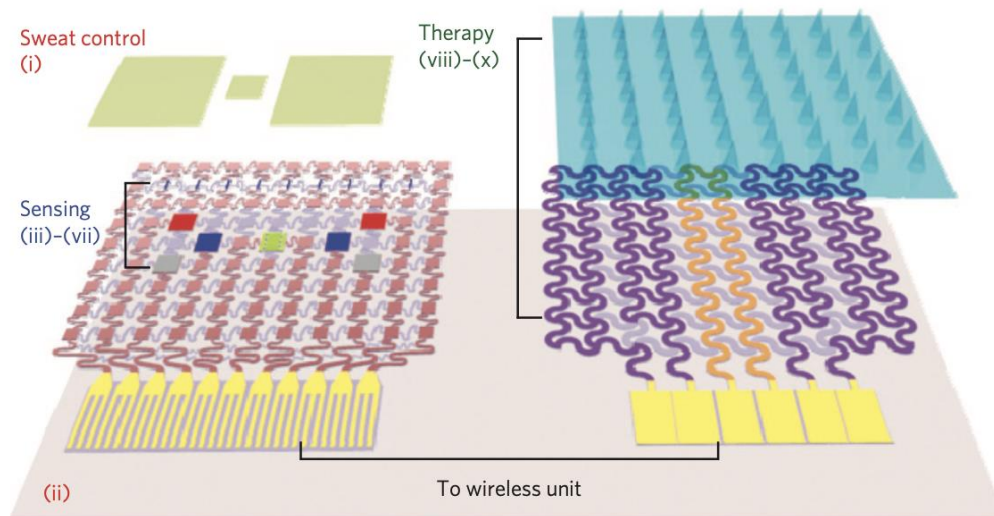


High-Density Silicon Array Patch

- High-density silicon array (~ 9500 needles/cm²) providing superior mechanical strength and 3× the active surface area
- Incorporates a ferrocene redox mediator to operate at a low potential, suppressing interference
- Functions purely as a single sensing channel (lacks addressability and therapeutic capabilities)



- 1 Patterning: photolithography
- 2 Needle shaping: a three-step Bosch deep etch
- 3 Metallisation: sputtering on an adhesion layer
- 4 Insulation and reference
- 5 Biofunctionalisation: self-assembled monolayer and mediator



Thermoresponsive Graphene Patch

- Soft, stretchable, and transparent graphene-based patch
- Integrates a multiplexed sensor suite (glucose, pH, humidity, temperature, tremor) and a therapy module
- Closed-Loop Delivery: an embedded heater melts a phase-change coating ($>41^{\circ}\text{C}$), dissolving bioresorbable needles to release Metformin

- 1 Structural base: SU-8 on a sacrificial layer**
- 2 Serpentine mesh: evaporation and wet etching**
- 3 Active layer: CVD graphene transfer and doping**
- 4 Release and transfer printing: surface micromachining**
- 5 Sensor functionalisation: selective electrodeposition**
- 6 Microneedles: replica moulding and UV curing**

Reliability & Takeaways

2018

Si hollow needle

- dominant failure mode: loss of insertion (~2 h limit)
- reference-electrode potential and enzyme stability > 4 days
- Not yet tested under mechanical stress or real daily use

2022

Si dense array

- 95% response after 5 reuse cycles (but drops to 80% after 10)
- Optimum at pH 7.4 and 35°C ($\approx$ skin temperature)
- No bleeding or inflammation in vivo (mice)

2016

Graphene patch

- Paired sensors: monitoring continues if one of a pair fails
- Signal stable for 6 hours without recalibration
- Temperature sensor prevents overheating during drug release

2025

Integrated arrays

- 93.6% in Clarke Zone A in vivo (rats)
- 2 of 4 needles survived $\rightarrow$ redundancy proven in practice
- Insulin dose $1.04 \text{ U} \pm 6.7\%$ $\rightarrow$ reproducible
- Batch consistency $\sim 95\%$; interference $< 4\%$

Common Gap

All four are short-term demonstrations: extended in-vivo validation, manufacturing scale-up and regulatory approval all remain ahead.